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November 20, 1973
SUPERSEDING
Fed. Spec. 00-I-675E
September 9, 1970

FEDERAL SPECIFICATION

# IRONERS, FLATWORK, COMMERCIAL (AND ACCESSORIES)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers chest-type flatwork ironers, ventilating canopies, folding machines, combination spreader-feeder machine, crossfolders, and stacker.

1.2 Classification.

1.2.1 Sizes. Flatwork ironers shall be of the following sizes, as specified (see 6.2):

Size 2 - 6 rolls (see table I).
Size 3 - 8 rolls.

1.2.2 Accessory items. Accessory items shall be of the following items, types, styles, sizes, and models specified (see 6.2):

Item 1 - Ventilating canopy (see 3.7.1).
Item 2 - Crossfolder (see 3.7.2).
Item 3 - Folding machine (see 3.7.3).

Type I - Large and small piece flatwork (see 3.7.3.1).

Style A - Single lane.
Style B - Automatic adjusting two lanes.
Style C - Automatic adjusting four lanes.

Type II - Small piece flatwork (see 3.7.3.2).

Item 4 - Folding machine with built-in crossfolder (see 3.7.4).

Style A - Single lane.
Style B - Multi-lane.

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Item 4A - Combination folder (multi-lane), crossfolder (multi-lane) and stacker (see 3.7.5).
Item 4B - Combination folder (single-lane), crossfolder (single-lane) and stacker (see 3.7.5).
Item 5A - Combination spreader and feeding machine (see 3.7.6).

Size 1 - (See 3.7.6.1).
Size 2 - (See 3.7.6.2).

Item 6 - Stacker, flatwork, small piece (see 3.7.7).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

CC-M-636 - Motor, Alternating-Current, (Fractional Horsepower).
CC-M-641 - Motor, Alternating-Current, (Integral Horsepower, 200 Hp and Smaller).

Federal Standard:

FED-STD-123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.



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Military Specifications:

| | | |
|---|---|---|
| MIL-T-152 | - | Treatment, Moisture and Fungus Resistant of Communications, Electronic, and Associated Electrical Equipment. |
| MIL-L-3153 | - | Laundry and Dry Cleaning Machinery and Equipment (For Fixed Installations), Preparation for Delivery Of. |
| MIL-P-43219 | - | Paddings and Covers for Flatwork, Ironer Roll. |

Military Standards:

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - | Marking for Shipment and Storage. |
| MIL-STD-461 | - | Electromagnetic Interference Characteristics, Requirements for Equipment. |
| MIL-STD-462 | - | Electromagnetic Interference Characteristics, Measurement Of. |

**(Copies of Military Specifications and Standards required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)**

LAWS AND REGULATIONS

29 CFR 1910 - OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA) STANDARDS

(The code of Federal Regulations (CFR) and Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Electrical Manufacturers Association (NEMA) Standard:

ICS-1970 - Industrial Controls and Systems

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)



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Underwriters' Laboratories (UL), Inc., Standards:

(Application for copies of applicable standards should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 1285 Walt Whitman Road, Melville, L.I., NY 11749; 1655 Scott Boulevard, Santa Clara, CA 95050.)

National Fire Protection Association (NFPA) Standard:

No. 70 - The National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)

American Welding Society (AWS) Standard:

Qualification Procedure of the American Welding Society

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, NY 10017.)

American Society of Mechanical Engineers (ASME) Standard:

Boiler and Pressure Vessel Code, Section VIII, Unfired Pressure Vessels

Welding Qualifications of the American Society of Mechanical Engineers

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.3 and 6.5).

3.2 Standard product. The ironers and accessories delivered under this specification shall be the manufacturer's commercial product, except for any changes necessary to comply with specification requirements. All like items furnished on any one contract, including parts and subassemblies thereof shall be new and interchangeable.



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3.3 Codes and standards. The ironers and accessories shall comply with the applicable requirements of NEMA ICS-1970, UL, NFPA No. 70, and OSHA 29 CFR 1910.

3.3.1 Compliance. Prior to approval of the first article, if one is submitted, or prior to commencing production, the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that the ironers and accessory item that he proposes to furnish under this specification meet the applicable requirements of NEMA ICS-1970, UL, NFPA No. 70 and OSHA.

3.3.1.1 NEMA. Acceptable evidence of meeting the requirements of NEMA shall be a certificate of compliance certifying that motor controllers, switches, relay and time delays conform to NEMA ICS-1970.

3.3.1.2 UL. Acceptable evidence of meeting the requirements of UL shall be a certificate of compliance certifying that the wiring, fittings and conduit conform to the applicable UL standard for use in non-hazardous locations.

3.3.1.3 NFPA. Acceptable evidence of meeting the requirements of NFPA shall be a certificate of compliance certifying that ironers and accessories wiring conforms to NFPA No. 70.

3.3.1.4 OSHA. Acceptable evidence of meeting the applicable safety and health standards of OSHA shall be a certificate of compliance certifying that the ironers and accessories furnished conform to OSHA 29 CFR 1910 264 (ii) (a).

3.4 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacture in his standard commercial ironer or accessory provided that the completed item complies with all provisions of this specification.

3.4.1 Pipe, tube, fittings, traps, and valves. Pipe, tube, fittings, traps, and valves used for air, steam, and condensate lines shall comply with accepted plumbing practice.

3.4.2 Conduit, fittings, and electric wire. Electrical conduit shall be electrical metallic tubing or flexible metallic conduit. All electric wire shall be stranded copper, heat-resistant grade, thermoplastic insulated. Solid (nonstranded) wire shall not be used.



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3.4.3 Motors. Motors shall conform to the requirements of CC-M-636 or CC-M-641, as applicable, and shall conform to the following requirements:

(a) All motors shall have windings impregnated to resist moisture.

(b) Each motor shall have the proper starting characteristics, ample power, and adequate thermal dissipation capacity to comply with requirements specified herein, with a reasonable factor of safety, commensurate with the application and expected duty cycle to which each motor might be subjected under normal operating conditions.

(c) Motors shall be equipped with ball bearings, except that motors of 1/2 horsepower (hp) or less when used in horizontal applications may have sleeve bearings.

(d) Motor bearings shall be of the permanently lubricated type or shall have adequate and accessible means for lubrication.

(e) Each motor of 1/8 hp rating or larger shall be provided with thermal-overload protection incorporated as a part of the motor or installed in the controller.

3.4.4 Motor control circuit. Each motor on ironers and accessory items, as applicable, shall be provided with a controller of the magnetic across-the-line type having a separate pole for each ungrounded conductor. Controllers for direct current motors of more than 1/2 hp shall be of the reduced voltage type having one or more steps of armature resistance for smooth acceleration. Pushbutton control stations shall be mounted at points convenient to the operator. Pushbuttons shall be protected or recessed to prevent accidental operation. Timers, limit switches, remote control pushbutton switches, signals, and indicating lights shall be provided to perform the intended functions.

3.4.5 Belts and pulleys. V-belts shall be of the grade commercially known as multiple-drive type except where a single belt is used with a fractional-horsepower motor, a V-belt commercially known as a fractional-horsepower belt may be supplied. Pulleys for V-belts shall be suitably grooved and shall not be a die cast type when used in powered drive. Chain type belts shall be of the silent or roller-chain type. Belt and chain tension shall be adjustable.



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3.5 Design and construction. The completely assembled items shall show that the assembly of components as applicable to the particular item has: (a) complete and proper interrelationship; (b) moving parts evidence no overheating; and (c) starting and stopping devices operate smoothly and positively. These requirements shall be determined when the equipment is tested as specified in 4.7.2.

3.5.1 Electrical characteristics. The electrical parts of the machines shall be suitable for the voltage, frequency, and number of phases for alternating current (AC) and voltage for direct current (DC) as specified (see 6.2).

3.5.2 Wiring and instructions. Wiring shall be installed in conduit which shall be provided between all components mounted on the equipment. Wiring shall be installed with materials specified in 3.4.2, as applicable, except metallic armored cable may be used where required to allow for belt, or other drive adjustment. All wiring shall terminate in a connection box with provisions for connections beyond the machines. Wiring and connection diagrams, including instructions for connecting controllers to be mounted separately from the equipment shall be furnished with the end item.

3.5.3 Safety devices. Belts, ribbon feeds, chains, shafts, pulleys, and gears shall be enclosed or guarded to prevent operator injury. Guards shall be cast iron or sheet metal. All parts of the guard shall be removable without disassembling pipes or fittings on the machine. There shall be access openings with covers over parts requiring adjustment or lubrication.

3.5.4 Equipment base. When specified (see 6.2), the equipment base shall be provided with a leveling device. There shall be provisions for anchoring the base to the floor, and anchor bolts shall be furnished.

3.5.5 Working pressure. Cylinders, chests, and chambers using steam shall be designed for a working pressure of 125 pounds per square inch gage (psig) steam and shall be capable of withstanding a hydrostatic pressure established in accordance with the provisions of the ASME Unfired Pressure Vessel Code when tested as specified in 4.7.1.

3.5.6 Gears and sprockets. Gears and sprockets shall have machined teeth. The use of die-cast gears or sprockets shall not be permitted on any powered drive. Gears shall be of ferrous or aluminum bronze material and sprockets shall be of ferrous metal. Noise reducing material of phenolic laminate may be used for drive and intermediate pinions. All pinions shall have faces at least as wide as the gears with which they mesh.



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3.5.7 Bearings and lubrication. Removable, aluminum bronze sleeve, ball, or roller bearings shall be provided for all shafts, rolls, and studs. Bearings shall be replaceable and shall be permanently lubricated unless oil cups or pressure grease fittings are provided. Gears and chains shall be oil or grease lubricated. Seals shall be provided to prevent the lubricant from entering parts of the machines where clothes are handled. Lubrication instructions including location, type of lubricant, and time intervals shall be indented or embossed on a noncorrodible metal plate and permanently affixed in accordance with the manufacturer's practice.

3.5.8 Piping. Integral piping, fittings, condensate traps, pressure reducing valves for air and steam, and drains shall be installed or furnished as complete assemblies suitable for installation with unions or other standard fittings. Each condensate trap shall be provided with a valve and by-pass.

3.5.9 Valves. Inlet steam and air supply connections shall be provided with valves. Condensate traps (with valve and by-pass) and steam, air drain, and other operational valves shall be accessible for operation and maintenance.

3.5.10 Spacing and extent of fasteners. The spacing and extent of welds, rivets, bolts, and screws shall insure fastening and prevent bulging of fastened metals.

3.6 Ironers. The flatwork ironers shall be chest type and shall have the number and minimum length of rolls specified in table I. Ironers shall be designed to operate on 125 psig steam pressure and at a temperature of 345°F. Ironers shall be capable of ironing sheets dry (not more than 7 percent moisture content) when the sheets are saturated and the total weight of the sheets are not more than 1.45 times the dry weight of the sheets and operating at standard speeds as specified in table I. These requirements shall be determined when the ironer is tested as specified in 4.7.3.1.

TABLE I. Roll dimensions and quantity and speed range

| Size | Minimum roll diameter (inches) | Quantity of rolls | Minimum roll length (inches) | Range of standard operating speeds (linear feet per minute) |
|---|---|---|---|---|
| 2 | 13-1/4 | 6 | 120 | 24-72 |
| 3 | 13-1/4 | 8 | 120 | 30-90 |



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3.6.1 Frames. Two ferrous metal frames with legs shall support the padded rolls, steam chests, and control mechanism at the operating level. The frame shall be connected with rods or structural shapes of ample strength to support the ironing components and prevent distortion of support members.

3.6.1.1 Padded rolls. The padded rolls shall be padded ferrous tube or pipe with ferrous metal heads and shall be driven through gears or sprockets and roller chains. The rolls shall operate in bearing mounts with springs or pneumatic or hydraulic cylinders arranged to exert the ironing pressure. The diameter of the rolls over the covering and padding shall be not less than as specified in table I.

3.6.1.2 Padding and covering. Padding shall conform to the requirements of MIL-P-43219, or when specified (see 6.2), spring padding shall be furnished. When flatwork ironer rolls are equipped with spring padding, there shall be a vacuum attachment for exhausting moisture from the padded rolls. When specified (see 6.2), a vacuum attachment for exhausting moisture from type I and II rolls shall be provided.

3.6.1.3 Steam chest. Steam chests shall be ferrous metal. Chests shall have a smooth machine finish with ground and polished ironing surfaces. For each padded roll, the ironer shall have one steam chest anchored to the frame. The chests shall be concave on top to suit the padded rolls, and flat or convex on the bottom. The last chest shall have a continuous radius rear section over which the goods shall pass to reach the bottom side of the chests on return type ironers. When specified (see 6.2), the underside of the chest shall be insulated.

3.6.1.4 Shafts. Ferrous metal shafts shall be provided for mounting of pulleys, gears, and rolls.

3.6.1.5 Feed. The feed shall consist of ribbons traveling over a power driven roll and a stationary plate member fitted with guiding pins or metal fingers. The ribbons shall be endless, 2-1/2 to 3 inches wide, of heavy feeder ribbon stock, sufficient in number to suit the width of the machine, and arranged to carry the goods to be ironed to the first padded roll. There shall be a padded finger roll of sufficient weight placed on the feeder ribbons to carry goods being ironed to the first padded ironing roll, when a vacuum type spreader, or automatic feeder spreader is being used. The finger roll shall rest on the feed ribbons and be powered by the motion of the feed ribbons. Means shall be provided for tightening the ribbons, and the drive chain, if used. The feed device shall be supported by ferrous metal brackets bolted to the front of the machine.



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3.6.1.6 Feed finger guard. The operator's metal finger guard shall extend across the entire width of the ribbons, ahead of the first padded roll and the steam chest. Movement of the guard in the direction of feeding, as a result of the operator's finger(s) passing between the guard and the ribbon feed, or bulging of material shall automatically stop the machine (see 4.7.2). The guard shall be located so the operator's fingers cannot pass between the guard and the ribbon feed. The height of the guard shall be sufficient to prevent the operator reaching over it and getting his fingers caught between the chest and the first padded roll.

3.6.1.6.1 Roll guard. When specified (see 6.2 and 6.3), the padded rolls shall be guarded so that the operator or any other person cannot reach into the rolls without removing the guard. This may be either a vertical guard on all sides or a complete cover. If a vertical guard is used, the distance from the floor or working platform to the top of the guard shall be not less than six feet.

3.6.1.7 Apron or ribbons. The upper and lower metallic-laced duck aprons or ribbons operate over a series of rotating rolls. The aprons or ribbons shall receive the goods being ironed after leaving the last roll and carry them against the bottom of the chests to the feed side where the lower apron or ribbons shall convey them back to the delivery side. Ribbons shall be 2-1/2 to 3 inches wide and in sufficient number to suit the width of their respective rolls.

3.6.1.7.1 Apron duck. Apron duck shall be No. 5 standard duck or heavier.

3.6.1.7.2 Apron or ribbon rolls. These rolls, whether for drive or support purposes, shall rotate in bearings. The rolls shall be turned from sturdy steel tubing, be equipped with secured heads having trunnions of high grade steel and be capable of maintaining permanent alinement. The rolls shall produce a uniformly distributed pressure over their entire length.

3.6.1.7.3 Apron release. Means shall be provided on apron flatwork ironers for simultaneously releasing both upper and lower aprons or ribbons from the ironing position without changing the initial tension. Aprons shall be automatically released with a roll pressure release (see 4.7.2).

3.6.1.7.4 Apron tension adjustment. Ironers shall have individual takeup devices for controlling the tension of each apron or ribbons. Each takeup device shall have a visible graduated scale so that the amount of adjustment may be determined and thus prevent the possibility of uneven stretching of the apron or ribbons.



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3.6.1.7.5 Apron tension rolls. Apron tension rolls shall be provided with a raised track in the form of a double spiral, arranged to prevent buckling and to keep the apron taut sidewise.

3.6.1.7.6 Apron guide rolls. Guide rolls shall be arranged to keep the aprons or ribbons properly aligned automatically.

3.6.1.8 Pressure device. The pressure device shall exert uniform pressure on the springs at both ends of the padded rolls and operate in conjunction with means for applying or releasing the tension on the aprons or ribbons on ironers, as applicable. Application and release of pressure shall be through mechanical, pneumatic, or hydraulic devices. Devices shall operate manually when power is not available. A pressure indicator for indicating the relative amount of pressure and a speed indicator showing the approximate peripheral speed of the rolls in feet per minute shall be mounted so they are readable from the feed side of the ironer.

3.6.1.9 Guides. Heat-resisting strings or tapes operating over spools shall be provided for guiding the laundry under the rolls. A string or tape rod, or other suitable and equivalent means, shall be fitted at the rear of the machine and shall be adjustable as to its location with respect to the last chest or padded rolls. Means shall be provided for automatically tightening the strings or tapes.

3.6.1.10 Pipe connections. Individual steam supply and condensate return pipes shall connect each steam chest to manifolds for steam supply and condensate return.

3.6.1.11 Electrical equipment. The ironer shall be electric-motor operated with pushbutton control. Electrical equipment shall conform to the general requirements specified herein.

3.6.1.12 Motors, controls, and operating speeds. The motor shall be started by a push button located at the front of the ironer and stopped by push button located both at the front and rear of the ironer. Motors shall be furnished with a mechanical speed changer to provide infinite delivery speed ranges from 1/3 to full speed. Means shall be provided wherein the mechanical speed changer can be changed from full speed to 1/3 speed (without stopping the ironer) (see 4.7.2). Provisions shall be made so that the speed changer may be set at low speed prior to starting the ironer. The mechanical speed changer controls shall be mounted on a metal bracket on the ironer or on a metal floor stand at operating level at the feed end of the ironer. A detachable hand crank shall be provided for operating the drive in the reverse direction. Means shall be provided for operating the machine forward at a slower speed for repadding purposes.



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3.6.1.13 Delivery table. A delivery table shall be furnished except when the ironer is to be used with a folding machine. The delivery table, shall have a wood top and metal base and shall be arranged so that the top may be adjusted from 27-3/4 inches minimum to 33-3/4 inches maximum height.

3.7 Accessories.

3.7.1 Item 1, ventilating canopy. When specified (see 1.2.2), a ventilating canopy shall be furnished with the flatwork ironer. The canopy shall be designed to fit on top of the flatwork ironer and shall be of the enclosed type, having hinged sides and ends that are opened by hand, or of the power lift type, as specified (see 6.2). The canopy shall completely cover all rolls of the flatwork ironer and shall be furnished with an exhaust fan or exhaust collar as specified (see 6.2). The power lift canopy shall open and close or raise the side and end panels as a unit from the top of the ironer by a power operated device and the hand operated type shall have side and end panels that open and close or raise with a counterbalanced device (see 4.7.2).

3.7.1.1 Side and end panels. The side and end panels shall have safety type panes to provide an unobstructed view of the interior of the canopy when the panels are closed. The adjustable panels shall rest against strips attached to the ironer when closed, and when open shall be horizontal or higher.

3.7.1.2 Exhaust fan. The exhaust fan shall be driven by a constant speed motor with thermal overload protection and with a manually operated controller.

3.7.2 Item 2, crossfolder. The crossfolder shall be for use in conjunction with a sheet folder designed to make two lengthwise folds. The machine shall be capable of producing three crossfolds to the lengthwise folded sheets. These requirements shall be determined when the machine is tested as specified in 4.7.3.2. The sheets shall be conveyed along the top of the crossfolder by means of a multiple belt conveyor. When the sheet reaches a predetermined position, an air blast shall push the sheet through the rollers folding the sheet in half, and at the same time delivering the sheet onto a lower conveyor running at 90 degrees to the upper conveyor. The second and third crossfold shall be formed on the lower conveyor, and the sheet shall be delivered to the operator at the side of the machine. The frame shall be fabricated of ferrous metal, welded to form a rigid structure.

3.7.2.1 Drive. The crossfolder shall be self-contained and furnished complete with motor(s), overloads for motor(s), driving mechanism, starters, relays, potentiometers for control circuits, fuses, and necessary terminals.



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3.7.2.2 Power transmission components. Ferrous metal shafts shall be provided for mounting pulleys, sprockets, gears, and rolls. Rolls shall be fabricated of seamless or welded ferrous metal tube either galvanized, chromium-plated, or covered with duck or rubber composition. Bearings shall be in accordance with the manufacturer's current commercial practice. Where flange ball bearings are used, they shall be sealed and self-aligning. Conveyor belts shall be adjustable.

3.7.2.3 Delivery table. The crossfolder shall have all the necessary delivery table(s) and shall be equipped for either right or left hand delivery, as specified (see 6.2). The table top height shall be not less than 30 inches from the floor.

3.7.2.4 Indicating counter. The crossfolder shall have a piece counter to record the number of flatwork pieces which have passed through the machine.

3.7.2.5 By-pass mechanism. The crossfolder shall automatically by-pass any sheet which has been delivered from the folder that is over 30 inches wide, or a sheet that is badly skewed which could jam the rollers (see 4.7.2). There shall be a sling or table to receive the automatically or by-passed sheets.

3.7.2.6 Quality grading controls. When specified (see 6.2), the crossfolder shall be equipped with quality grading controls whereby the operator of an ironer can warn the operator of the crossfolder by means of lights or audible signals that a piece of unsatisfactory quality (torn, stained, etc.) is approaching and should be removed (see 4.7.2).

3.7.2.7 Air supply. The crossfolder shall have an air supply inlet capable of receiving air at a minimum pressure of 100 psig. The air inlet shall have a pressure reducing valve with a gage to insure constant pressure, a filter to insure clean air, and when required (see 6.2), a lubricator to lubricate the air equipment.

3.7.3 Item 3, folding machine. The folding machine shall have a ferrous metal frame with rods or structural shapes for alignment and shall be equipped with not less than four casters or anchor bolts in the base, as specified (see 6.2 and 6.4). The machine shall have a tension adjustment for the conveyor ribbons. The rolls shall be fabricated from seamless tube, welded ferrous metal tube, non-ferrous metal, or solid bar either galvanized or chromium-plated or covered with duck or rubber composition and shall be mounted on fully enclosed, lint proof, ball or roller bearings. The lead top roller shall have a guard extending the entire length of the roll to prevent fingers from insertion.



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3.7.3.1 Type I, style A, B, and C. The type I, style A, B, and C folding machines shall be for automatically folding large and small piece flatwork and shall quarterfold the items lengthwise and deliver the folded item on to a delivery table or crossfolding machine. There shall be an indicating counter furnished for each lane to record the number of folded flatwork pieces which have passed through the machine (see 4.7.2). When specified, the machine shall make at least one primary fold by means of an air blast (see 6.2). The single lane machine shall be designed to fold flatwork from 30 inches to 120 inches wide and from 30 inches to 108 inches long when fed across the 120 inch width of the folder. The two and four lane machines shall be adjustable to two lanes of 60 inches wide and to four lanes of 30 inches wide, respectively, and from 30 inches to 100 inches long. These requirements shall be determined when the folding machine is tested as specified in 4.7.3.3.1.

3.7.3.1.1 Driving means. The folder shall be self-contained and shall include an independent power source.

3.7.3.1.2 By-pass mechanism. The machine shall be equipped with a by-pass mechanism designed to permit work to pass through the folder without being folded, and deliver it directly to a table (see 4.7.2).

3.7.3.1.3 Interlocking safety device. The folding machines furnished shall be capable of being interlocked electrically with an ironer so that in the event a rolled or bundled item approaches the folder either the feeding operators or the receiving operator can stop both machines simultaneously (see 4.7.2). The folding machine furnished shall be compatible for operation with its intended ironer use.

3.7.3.1.4 Static electricity. The machine shall be equipped with a static electricity eliminator capable of dissipating all conditions of static accumulation encountered while processing any type of fabric.

3.7.3.1.5 Delivery table, accessories, and ribbons. The folder shall be supplied complete with all necessary ribbons, accessories, and delivery tables as required.

3.7.3.1.6 Automatic measurement of work. A device shall be provided which shall automatically measure the length of the piece and vary the operation of the folding device in relation to the length of the piece being folded (see 4.7.2). When required (see 6.2), the control shall incorporate voltage stabilizing means to prevent service voltage fluctuation causing improper folding.

3.7.3.2 Type II. The type II folding machine shall be designed for automatically folding and stacking small piece flatwork (hand towels, bath towels, napkins and pillow cases). The machine shall fold the items into sixths; twice widthwise (in thirds) and once (in half) lengthwise. Unless otherwise specified (see 6.2), the machine shall be provided with a selector that will permit the operator to select the number of items to be counted and stacked in increments of 10, 15, 20, and 25 pieces per bundle. These requirements shall be determined when the folding machine is tested as specified in 4.7.3.3.2. The machine shall be movable and equipped with casters.



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3.7.3.2.1 Driving means. The folders shall be self-contained, including individually motor-driven components except for the compressor, if compressed air is utilized.

3.7.3.2.2 Conveyor. When specified (see 6.2), the folding machine shall be equipped with a conveyor. The length of the conveyor shall be as specified (see 6.2).

3.7.4 Item 4, folding machine, style A and B, with built-in crossfolder. The folding machine with built-in crossfolder shall be capable of performing the functions of a primary folder, with two lengthwise folds and automatically providing three crossfolds (see 4.7.3.4). The folded articles shall be delivered to the operator at the side of the machine. The machine shall be style A, single lane or style B, multi-lane design. The machine shall be capable of instantly converting from crossfolding to an automatic folding machine (see 4.7.2). The machine shall be self-contained and shall include an independent power source. Style B machines shall be provided with either two or four folding lanes, as specified (see 6.2), and a single crossfolding lane. When specified (see 6.2), at least one primary fold shall be accomplished by means of an air blast.

3.7.4.1 By-pass mechanism. The machine shall be equipped with an automatic safety device that automatically by-passes the crossfolding mechanism, if an item is not properly quarter-folded, and discharges the item to the rear of the folding machine, or on, or into a catch apron or table.

3.7.4.2 Delivery table, accessories, and ribbons. The folder shall be supplied complete with all necessary ribbons, accessories, and delivery table or tables as required. The table top height shall be not less than 30 inches from the floor. The crossfolder shall have a right or left hand discharge (determined from operator's end) as specified (see 6.2). When specified (see 6.2), the machine shall be equipped with quality grading controls conforming to requirements as specified in 3.7.2.6.

3.7.5 Item 4A, and item 4B, combination folder, crossfolder, and stacker. The combination folder, crossfolder, and stacker shall be an integral unit, which shall be equipped with a conveyor feed. The machine shall conform to requirements specified in 3.7.4 through 3.7.4.2 except that final operation shall be automatic stacking, counting and automatic conveying of stacked work to a discharge conveyor or delivery table (see 4.7.3.5). Item 4A shall be multi-lane and item 4B shall be single-lane. Item 4A shall be provided with either two or four folding lanes, as specified (see 6.2), and a single crossfolding lane. When specified (see 6.2), items 4A and 4B shall have at least one primary fold accomplished by means of an air blast.



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3.7.6 Item 5A, combination spreader-feeder machines. The combination spreader and feeder machine shall be electric motor-driven and designed for use with any 120/126 inch flatwork ironer without the need to modify or rework either machine. The machine shall operate at surface speeds to 150 FPM (see 4.7.2 and 4.7.3.6). The machine shall receive manually fed, damp-dry, flatwork and automatically straighten, square, remove wrinkles and creases, and convey the flatwork into the receiving end of the flatwork ironer (see 4.7.3.8.1). The speed of travel of the flatwork through the machine shall be variable from at least 50 to 150 FPM. The speed control shall have a speed indicator, graduated in FPM or percent of maximum speed to properly synchronize the spreader-feeder conveyor with the ironer feed ribbons. The indicator reading at any setting shall be within 10 percent of the actual delivery speed of the machine (see 4.7.3.6). All electrical and mechanical controls required to operate the machine shall be in one location and accessible to the operator.

3.7.6.1 Size 1. Except for manual feeding of the first 12 to 20 inches at the midpoint of the flatwork selvage edge to the receiving end of the machine no further flatwork preparation or manual assistance shall be required to feed the machine or convey the flatwork into the ironer (see 4.7.3.8.1). An electric motor-driven forward and reverse traverse system shall be incorporated into the design. This system shall move the entire machine 4 to 5 feet away from the ironer, to provide access for manual feeding of small flatwork directly to the ironer (see 4.7.3.8).

3.7.6.1.1 Dimensions. The overall dimensions of the installed machine shall not exceed 135 inches wide, 68 inches high, and 134 inches long. The reciving station shall be not less than 40 nor more than 48 inches above the floor.

3.7.6.1.2 Frame construction. The framework of the machine shall be a bolted, riveted, or welded assembly of structural steel members. The supporting legs shall have at least 5 inches of vertical adjustment to permit alignment of the machine to a 120/126 inch flatwork ironer. Except for the forward and reverse traverse guides, which may be floor mounted, all parts and subassemblies shall be directly or indirectly mounted on the basic framework of the machine.

3.7.6.1.3 Drive train. The drive train for the conveyor system, and the forward and reverse traverse system shall be of a positive acting type that will preclude slippage in the traverse system, and insure that the speed ratio between rotating elements in the conveyor system remains constant at all conveyor speeds (see 4.7.3.7).



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3.7.6.2 Size 2. Unless otherwise specified (see 6.2), two or three loading stations shall be provided with each station equipped with an easily accessible set of loading clips. Each station shall have a switch to signal the control when the sheet is clipped. The machine shall be constructed so that any station can be operated independently of the other stations and be able to function with one or more operators. Sheets which are clipped at the loading station shall be delivered to a central location and transferred to devices which spread, center and deposit them on the spreader-feeder conveyor. The spreader-feeder conveyor conveys the sheets onto the feed ribbons of the flatwork ironer (see 4.7.3.8.1).

3.7.6.2.1 Dimensions. The overall dimensions of the installed machine shall not exceed 155 inches wide, 73 inches high and 84 inches deep.

3.7.6.2.2 Frame construction. Shall be a bolted, riveted, or welded assembly of structural steel members. The machine shall align with the feed ribbons of the flatwork ironer by means of leveling screws or conveyor adjustment. The machine shall be equipped with casters and floor lock brake so that it can be easily moved into and out of position in front of the flatwork ironer.

3.7.6.2.3 Drive train. Drive train shall consist of three motors to be used for drive purposes and to spread sheets as they are fed into the spreader feeder conveyor.

3.7.6.2.4 Controls and counter. Means shall be provided to direct sheets from the loading stations, in sequence, to the central position for delivery to the conveyor and to control sheet spacing on the ironer. Each loading station shall be equipped with a counter to determine the number of sheets being fed. The machine shall be provided with quality grading controls and shall operate in conjunction with grading controls provided in crossfolder and folder crossfolder whereby the operator(s) can signal that a piece of unsatisfactory quality is being processed.

3.7.7 Item 6, small piece flatwork stacker. The small piece flatwork stacker shall be self-contained and driven by a motor independent of the ironer power source and shall be designed to receive small pieces of flatwork from the delivery apron of an ironer and automatically stack the piece. The machine shall be capable of receiving and stacking from five lanes. The speed of the stacker shall be adjustable from a minimum of 60 feet per minute to 150 feet per minute. The conveyor ribbons shall be provided with a means of adjustment. Pieces being conveyed shall activate a means to neatly stack the pieces on the stacking bars. Through the use of electrical switches the machine shall be adjustable to stack pieces from 12 to 48 inches in length. These requirements shall be determined when the stacker is tested as specified in 4.7.3.9.



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3.7.7.1 Frames. Frames shall be fabricated from ferrous metal and held in alignment by ferrous metal rods or structural carbon steel members. The frame shall be inclosed by carbon sheet steel which shall be removable, when adjacent to an inclosed mechanism, for inspection and maintenance purposes.

3.8 Moisture and fungus-resistant treatment. When specified (see 6.2), electrical components of equipment covered by this specification shall be moisture and fungus-resistant treated with material conforming to and applied as required by MIL-T-152.

3.9 Electromagnetic compatibility. When specified (see 6.2), equipment shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461.

3.10 Finish and color. Unless otherwise specified (see 6.2), each item of equipment shall be finished according to the manufacturer's standard practice and as specified in 3.10.1 and 3.10.2. Emergency stop and normal stop push buttons shall be red, and in addition, feed guards shall be red and the red shall be limited to stop push buttons and feed guards. The color of the start push buttons shall be in accordance with the manufacturer's standard commercial practice.

3.10.1 Metallic coating thickness. When galvanizing or other protective metallic coating is used, the thickness of the coating shall be not less than 0.0005 inch thick, except that in places difficult to coat a thickness of 0.00025 inch shall be acceptable.

3.10.2 Exposed corrosion-resistant metals. Corrosion-resisting steel and plated metal shall be free from scale.

3.11 Identification marking. Each item of equipment and replacement part covered by this specification shall be marked for identification in accordance with the manufacturer's standard practice provided the items are legibly and permanently marked with the manufacturer's name or trademark of such known character that the source of manufacture may be determined and contains the type and size of the item, and the electrical characteristics, as applicable.

3.12 Equipment manual. Unless otherwise specified (see 6.2), for civil agency procurement, the supplier shall furnish with each flatwork, ironer or accessory, an equipment manual containing complete installation, operation and maintenance instructions, wiring diagrams, trouble-shooting chart and an illustrated repair parts section. The repair parts section shall include suppliers part number, description of parts, actual manufacturer's name and part number.



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3.13 Workmanship. The completely fabricated item shall be free from defects such as fractures, splits, punctures, tears, dents, creases, bows, miscasts, deteriorations, or malformations. There shall be no sharp edges, burrs, or slivers.

3.13.1 Application of finish. The finish supplied to the end item shall be continuous, smooth, adherent, have no imbedded foreign matter, and contain no sags, runs, drips, creeps, laps, bubbles, or streaks. No rust or toolmarks shall show through the coating.

3.13.2 Welding and brazing. All surfaces of parts to be welded or brazed shall be clean. Welding and brazing shall be accomplished in a manner to prevent the occurrence of burn holes, cracks, fractures, or incomplete fusion. All scale and flux deposits shall be removed from finished welds.

3.13.3 Soldering. Soldering shall be complete and adherent with all flux and residue removed and shall contain no pinholes.

3.13.4 Threaded fasteners. Threaded fasteners shall not be broken, stripped, fractured, or loose.

3.13.5 Rivets. Rivets shall be driven to completely fill the holes, with full size heads finished and in full contact with the surface of the riveted members, and shall not be loose.

3.13.6 Electrical wiring. Wiring shall not be cut, abraded, or have excess insulation stripped, and shall be securely joined at terminals. Wiring shall have slack to provide strain relief.

3.14 Welders. The welders performing the welding on pressure assemblies shall be certified welders and shall have passed the qualification test as prescribed by either the "Standard Qualification Procedures of the American Welding Society" or "Welding Qualification of the American Society of Mechanical Engineers" code. The supplier shall furnish the Government with a list of names of his welders who are certified and have passed the test of either referenced codes and he shall certify that the welding of this equipment was performed by welders on the list.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other



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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article. When a first article is required, the ironer and accessory item specified shall be examined for defects listed in table II, specified dimensions and tested as specified in 4.7.1 through 4.7.3.9, as applicable and 4.7.4, when applicable. Failure of any test or the presence of visual or dimensional defects shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 End item inspection. The inspection lot shall be all ironers of one size and accessories of specified type, style, size, item and model, as applicable, offered for inspection at one time.

4.4.2.1 Visual inspection. The ironer and accessory shall be examined for the defects in table II. The inspection level shall be II with a AQL (acceptable quality level) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.



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TABLE II. Classification of defects

| Examine | Defect | Classification | |
|---|---|---|---|
| | | Major | Minor |
| Finish | Not finished where required. | X | |
| | Ironer and accessories not color specified (if other than manufacturers standard). | | X |
| | Scale on corrosion-resistant metal or plated metal. | | X |
| | Emergency stop and normal stop push buttons not red. | X | |
| | Feed guards not red. | X | |
| | Color red used on other than stop push buttons and feed guards. | X | |
| | Not continuous, smooth, adherent, imbedded foreign matter, has sags, runs, drips, creeps, laps, bubbles, or streaks (painted components). | | X |
| | Rust or toolmarks. | | X |
| Construction | Any component missing or not specified type. | X | |
| | Any component split, puncture, tear, dent, crease, deteriorated, or malformed. | X | |
| | Sharp burr, sliver, or sharp edge. | X | |
| | Push button control stations not convenient to operator. | X | |
| | Push button not protected or recessed. | X | |
| | Exposed belts, chains, shafts, pulleys, or gears. | X | |
| | Any functioning component that requires abnormal force to operate. | X | |
| | Bearings not replaceable. | X | |
| | Guards not removable without disassembling pipes or fitting. | X | |
| | Base not in full contact with a level floor, (when applicable). | X | |
| | No provision for anchoring base to floor (when applicable). | | X |
| | Component not connected or joined as specified. | X | |
| | Pinion not as wide as meshing gear. | | X |
| | Seals not provided to prevent grease from entering parts of machine where flatwork is handled. | X | |



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TABLE II. Classification of defects (cont'd)

| Examine | Defect | Classification | |
|---|---|---|---|
| | | Major | Minor |
| Construction (cont'd) | No means to adjust belt and chain tension. | X | |
| | Condensate trap without valve and by-pass. | X | |
| Ironer | Not chest type. | X | |
| | Control mechanism not at operating level. | X | |
| | No leveling devices. | X | |
| | No springs to exert ironing pressure. | X | |
| | Not a steam chest for each padded roll. | X | |
| | Chests not concave on top or flat or convex on bottom. | X | |
| | Guard across line of feed missing or allows operator's fingers to pass through. | X | |
| | No apron release device. | | X |
| | Delivery table top not adjustable. | X | |
| Item 1 Ventilating canopy | All ironing rolls not covered. | X | |
| | Not outlet collar or exhaust fan, (as applicable). | X | |
| | Enclosed type side and end panels do not lift to horizontal position with counterbalanced device. | X | |
| Item 2 Crossfolder | Quality grading controls not as specified. | X | |
| Item 3 Folding machine | No bolt holes in base. | X | |
| | Not removable. | | X |
| | Finger guard on lead roll missing or permits finger insertion. | X | |
| Item 4 Folding machine with built-in crossfolder | Not specified lane design. | X | |
| | Not self-contained. | X | |
| | Does not include independent power source. | X | |
| | Not specified type discharge. | X | |
| | Finger guard missing. | X | |



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TABLE II. Classification of defects (cont'd)

| Examine | Defect | Classification | |
|---|---|---|---|
| | | Major | Minor |
| Construction (cont'd) | | | |
| Items 4A and 4B Folder, cross-folder, and stacker | Not an integral unit (defects as classified for item 4). | X | |
| Item 5A Spreader and feeder machine | Operator controls not grouped together. | X | |
| | Controls not at receiving end of machine. | X | |
| | Variable speed control dial or gage missing. | X | |
| | Basic frame not structural steel. | X | |
| | Supporting legs not adjustable. | X | |
| | Parts and subassemblies (except traverse guides not mounted directly to basic frame). | X | |
| | Not specified size. | X | |
| Item 6 Stacker (small piece) | Not an inclosed frame. | | X |
| Welding and brazing (where applicable) | Missing, incomplete, burn holes, fractured, or not fused. | X | |
| | Slag inclusion, undercut, not smooth and uniform, or deposit of scale and flux not removed. | X | |
| Soldering (where applicable) | Missing, incomplete, or not adherent. | X | |
| | Not clean (flux or residue not removed) or contains pinholes. | | X |
| Threaded fasteners | Broken, stripped, or fractured. | X | |
| Rivets | Broken, not peened, or not peened sufficiently. | X | |
| | Rivets loose. | | X |



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TABLE II. Classification of defects (cont'd)

| Examine | Defect | Classification | |
|---|---|---|---|
| | | Major | Minor |
| Construction (cont'd) | | | |
| Electrical apparatus wiring | Wiring cut, abraded, insulation stripped, not joined, or loose at terminal. | X | |
| | Not inclosed in specified conduit, where required. | X | |
| | Wiring not terminated in junction box. | X | |
| | Slack not provided for wiring to relieve strain. | | X |
| Piping, tubing, fittings and valves | Improperly connected or installed, (threads stripped, leaks). | X | |
| Gages and instruments | Not properly mounted or no means for adjustment. | X | |
| Lubrication | Not lubricated where required. | | X |
| | Excessive lubrications. | | X |
| | Fitting not accessible for lubrication. | | X |
| | Instructions missing or illegible. | X | |
| Marking identification | Missing, incomplete, illegible. | | X |
| Equipment manual | Missing, incomplete, not legible. | X | |

4.4.2.2 Dimensional examination. The ironer and accessory item shall be examined for compliance with dimensions specified. Any dimension not within this specification tolerance shall be a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.4.2.3 End item testing. When a first article is not required, the first produced ironer and accessory item specified shall be examined and tested as specified in 4.7.3.1 through 4.7.3.9, as applicable and when applicable 4.7.4. Each sample shall be tested as specified in 4.7.1 and 4.7.2. Failure of the unit to pass the applicable functional test(s) shall be cause for rejection of the lot. The inspection level shall be S-2 with an AQL of 2.5 defects, expressed as defects per hundred units.



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4.5 Code and standard examination. Certificate of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with specification requirements shall be examined and validated.

4.6 Examination of preparation for delivery. Examination of preparation for delivery shall be in accordance with MIL-L-3153.

4.7 Tests.

4.7.1 Hydrostatic pressure. Assemblies of cylinders and chambers that, in operation, contain steam shall be subjected to a hydrostatic pressure test determined in accordance with the provisions of the ASME Unfired Pressure Vessel Code based on a working pressure of 125 psig. The pressure shall be maintained for 10 minutes. Inability of the cylinders and chambers to meet the requirements of 3.5.5 shall constitute failure of the test.

4.7.2 Operational testing. The ironers and accessories shall be operationally tested by the supplier as follows, for a length of time sufficient to determine compliance with the requirements listed below. Each power operated item shall be connected to electric power of the specified voltage, frequency, and number of phases, air and hydraulic power (as applicable for operation), and tested as specified to determine that the applicable performance requirements have been met and all motors have ample power and operate without exceeding the permissible temperature rise. The following shall also be checked:

(a) The assembly of components, starting, stopping, and control devices, moving parts and bearings for compliance with the requirements of 3.5.

(b) Operation of the feed finger guard in type I ironers for compliance with 3.6.1.6.

(c) Operation of the apron or ribbon release for compliance with 3.6.1.7.3.

(d) Operation of the mechanical speed changer for compliance with 3.6.1.12.

(e) Operation of hand opening device or power lift of the ventilating canopy for compliance with 3.7.1.

(f) Operation of by-pass mechanism of crossfolder for compliance with 3.7.2.5.

(g) Operation of quality grading controls for compliance with 3.7.2.6.



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(h) Operation of indicating counter for type I folding machine for compliance with 3.7.3.1.

(i) Operation of by-pass mechanism for compliance with 3.7.3.1.2.

(j) Operation of interlock safety device for compliance with 3.7.3.1.3.

(k) Operation of automatic measuring device for compliance with 3.7.3.1.6.

(l) Operation of conversion mechanism for compliance with 3.7.4.

(m) Operation of combination spreader-feeder for compliance with 3.7.6.

Inability of the ironers and accessories to meet the applicable requirements of the specified paragraphs shall constitute failure of the test.

4.7.3 Functional.

4.7.3.1 Flatwork ironers. Connect the ironing machine to an electrical power source having characteristics as specified and to steam at 125 psig and a temperature of 345°F and operate the machine at speeds specified in table I. Iron 10 sheets containing moisture not more than 1.45 times the dry weight. Stop the machine and weigh the sheets. Inability of the flatwork ironers to meet requirements specified in 3.6 shall constitute failure of the test.

4.7.3.2 Crossfolder. Connect the crossfolder to electric power and air supply, and feed full size folded bed sheets through the machine for one hour and in conjunction with accessories of normal use. Inability of the crossfolder to meet the requirements specified in 3.7.2 shall constitute failure of the test.

4.7.3.3 Folding machines.

4.7.3.3.1 Type I folding machine. Connect the type I folding machine to the specified electric power source and fold flatwork ironing in sizes from 30 inches by 30 inches to 120 inches by 108 inches for one hour and then stop the machine. This test shall be performed in conjunction with accessories of normal use. Inability of the folding machine to meet the requirements specified in 3.7.3.1 shall constitute failure of the test.



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4.7.3.3.2 Type II folding machine. Connect the type II folding machine to the specified electrical power source and fold hand towels into increments of 10, 15, 20, and 25, respectively. Stop the machine and examine the fold and count the number of hand towels in each increment to determine compliance with the requirements specified in 3.7.3.2. Any nonconformance shall constitute failure of the test.

4.7.3.4 Folding machine with built-in crossfolder. Connect the machine to the specified electric power source and air supply and feed twenty full size bed sheets or one or more sheets through the machine for one hour. This test shall be performed in conjunction with accessories of normal use. Inability of the folding machine to meet the requirements specified in 3.7.4 shall constitute failure of the test.

4.7.3.5 Combination folder, crossfolder and stacker. Connect the machine to the specified electric power source and air supply and proceed as specified in 4.7.3.4. Inability of the combination folder, crossfolder and stacker to meet the requirements specified in 3.7.5 shall constitute failure of the test.

4.7.3.6 Conveyor speed. The machine shall be run sufficiently long at 50, 100, and 150 FPM, to compare the actual delivery speed with the speed indicator reading. Any discrepancy in excess of 10 percent between the actual speed and the reading of a surface speed indicator shall constitute failure of the test (see 3.7.6).

4.7.3.7 Conveyor system. For an 8 hour period the spreader-feeder shall be run continuously at 50, 100, and 150 FPM, for equal periods of time within the 8 hours. The delivery speed of the conveyor elements in FPM shall be checked at 1/2 hour intervals by means of a surface speed indicator. Any variation of speed ratio between elements at any speed cycle (50, 100, and 150 FPM), shall constitute failure to comply with this test (see 3.7.6.1.3).

4.7.3.8 Combination spreader and feeding machine. The forward and reverse traverse system shall be run through not less than 50 complete forward and reverse cycles. Any evidence of slippage, malfunction or traverse less than 4 to 5 feet shall constitute failure of this test (see 3.7.6.1).

4.7.3.8.1 Performance. The following shall be required to test performance:

(a) One or more machine operators, as required.

(b) Fifty (50) damp dry cotton bed sheets as received from a washer-extractor. The selvage edges of the bed sheets shall be between 115 and 120 inches long. With the machine operating at a surface speed of 50 FPM, the operator shall hand feed the first 12 to 20 inches at the midpoint of the selvage edge of a bed sheet into the receiving end of the machine, or shall attach sheets to clips at each loading station. Without further operator assistance, the bed sheet shall be permitted to pass through the machine.



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After the first sheet has passed the receiving end, 24 more bed sheets shall be similarly fed into the machine, in turn. The surface speed shall then be increased to 150 FPM and the remaining bed sheets fed into the machine in a like manner. Any evidence of the following shall constitute failure of the test (see 3.7.6.1 and 3.7.6.2).

1. Bed sheet does not travel completely through the machine without additional operator assistance.
2. Bed sheet is not delivered straight and square at the delivery end.
3. Any evidence of folds or creases at the point where the bed sheet leaves the machine.

4.7.3.9 Small piece flatwork stacker. Connect the small piece flatwork stacker to electrical power source and air supply when applicable and stack at least 50 pieces of flatwork at least 18 inches by 48 inches to determine compliance with the requirements specified in 3.7.7.

4.7.4 Electromagnetic compatibility tests. When electromagnetic compatibility is required, the first article or initial production unit, as applicable shall be tested by the supplier in accordance with test method CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the electromagnetic test report and the requirements meet MIL-STD-461.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Preservation, packaging, and packing shall be in accordance with levels A, B, and C requirements of MIL-L-3153 as specified (see 6.2).

5.2 Marking.

5.2.1 Civilian agencies. In addition to any special marking required by contract or order (see 6.2), interior packages and shipping containers shall be marked in accordance with FED-STD-123.

5.2.2 Military agencies. In addition to any special marking required by the contract or order (see 6.2), interior packages and shipping containers shall be marked in accordance with MIL-STD-129.



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6. NOTES

6.1 Intended use. The flatwork ironers and accessory items are intended to be used in fixed laundry installations.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size flatwork ironer required (see 1.2.1).
(c) Accessory item required (see 1.2.2).
(d) First article (see 3.1, 4.2 and 6.5).
(e) Electrical characteristics required (voltage, phase, cycles) (see 3.5.1).
(f) When a leveling device is required (see 3.5.4).
(g) When a vacuum exhaust attachment is required for type I and II ironer (see 3.6.1.2) and when spring padding is required.
(h) When underside of chest is to be insulated (see 3.6.1.3).
(i) When padded rolls are to be guarded (see 3.6.1.6.1).
(j) Type of ventilating canopy, when a canopy is required (see 3.7.1).
(k) Whether an exhaust fan or exhaust collar for ventilating canopy is required (see 3.7.1).
(l) Whether right or left hand delivery is required for crossfolder (see 3.7.2.3).
(m) When quality grading controls are required (see 3.7.2.6 and 3.7.4.2).
(n) When a lubricator is required for crossfolder (see 3.7.2.7).
(o) When casters or bolt holes are required (see 3.7.3).
(p) When air blast for primary fold(s) is required (see 3.7.3.1).
(q) When voltage stabilizing means is required for type I folder (see 3.7.3.1.6).
(r) When different stacking capability is required (see 3.7.3.2).
(s) Length of conveyor when required (see 3.7.3.2.2).
(t) Number of folding lanes and whether air blast folding is required (see 3.7.4).
(u) Whether crossfolder delivery is right or left hand (see 3.7.4.2).
(v) Number of folding lanes for item 4A and whether air blast is required (see 3.7.5).
(w) Number of loading stations if otherwise specified (see 3.7.6.2).
(x) When moisture and fungus-resistant treatment is required (see 3.8).
(y) When electromagnetic compatibility is required (see 3.9).
(z) Finish when other than specified (see 3.10).



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(aa) When drawings are required for installation purposes, request that successful bidder furnish drawings showing the exact dimensions of the item, overall floor space requirement, location of holddown bolts, location of inlet and outlet lines, point where electrical connections are to be made, and the electrical characteristics of the motor and any electrical equipment. Air Force purchases require drawings within 10 days after award to the successful bidder.

(bb) When the supplier is to furnish supervision of installation of demonstration service of items. If required, the location, approximate date, the person or office to report to, and any accommodations or services to be supplied by the installation or agency shall be stated.

(cc) When equipment manuals are required for civil agencies (see 3.12), and for military procurements (see 6.2.1).

(dd) Levels of preservation, packaging, and packing required (see 5.1).

(ee) Special marking when required (see 5.2.1 and 5.2.2).

6.2.1 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

6.3 Flatwork ironers. The requirements for the padded rolls to be guarded is deemed necessary for mental institutions (see 3.6.1.6.1).

6.4 Flatwork folders. When a folder is ordered for an existing ironer (see 3.7.3) the name, size, model, and serial number of the existing ironer should be specified; it is also advisable to require installation by the crossfolder manufacturer.

6.5 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.



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6.6 Successful commercial application. Invitation for bids should specify that no item of equipment shall be acceptable unless the manufacturer has had equipment of approximately the same type and design as that specified operating successfully in a commercial or institutional laundry or laundries for at least one year. Equipment installed for test purposes in a manufacturer's plant or laboratory shall not come within the category of successful commercial application.

6.7 Deletions. Deleted ironer and accessories no longer required by military or civil agencies.

| | |
|---|---|
| Flatwork Ironer | Size 1 |
| Accessories: | |
| Item 5 | Spreading machine<br>Model A<br>Model B |
| Item 7 | Straightening, smoothing, and feeding machines<br>Type I<br>Type II |

MILITARY CUSTODIANS:

Army - GL
Navy - YD
Air Force - 84

Review activity:

Army - MD

User activities:

Army - CE
Navy - MC, MS

Preparing activity:

Army - GL

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
VA-DMS

Project No. 3510-0202

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